

Solubility of Gases

Gases dissolve in water too — and the dissolved oxygen that keeps fish alive shrinks as water warms.

Gases dissolve in liquids

Gases also dissolve in water. **Oxygen** dissolves only to a small extent, but this **dissolved oxygen sustains all aquatic life** (fish, plants and other organisms). Gas dissolved evenly in water makes a **uniform mixture**.

Effect of temperature on gas solubility

Opposite to solids! The solubility of **gases generally decreases as temperature increases**.

- **Cold water** holds **more** dissolved oxygen → good for aquatic life.
- **Warm water** holds **less** oxygen → can harm fish (a reason warming water/thermal pollution is dangerous).

Everyday clue: a cold fizzy drink fizzes more when it warms up — the dissolved gas escapes as solubility drops.

Where students lose marks

Trap 1 — Gas solubility DECREASES with temperature (the opposite of most solids). Don't mix them up.

Trap 2 — Cold water holds MORE oxygen. So aquatic life is better supported in cooler water.

Trap 3 — Dissolved oxygen is small but vital. Even a little dissolved O_2 keeps fish alive.

Model answers — write them like this

Example A. Why can fish survive better in cold water than in very warm water?

Step 1: Gas solubility decreases as temperature rises.

Step 2: Cold water holds more dissolved oxygen for the fish to breathe.

∴ More dissolved oxygen in cold water supports aquatic life.

Example B. Why does a fizzy drink go flat faster when warm?

Step 1: The drink contains dissolved carbon dioxide.

Step 2: Warming lowers gas solubility, so the gas escapes.

$\therefore$ Less gas stays dissolved when warm, so it goes flat.

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